Simple Eigenvalues and Non-vanishing Eigenvectors of Random Matrices

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Motivation

The Anderson Model

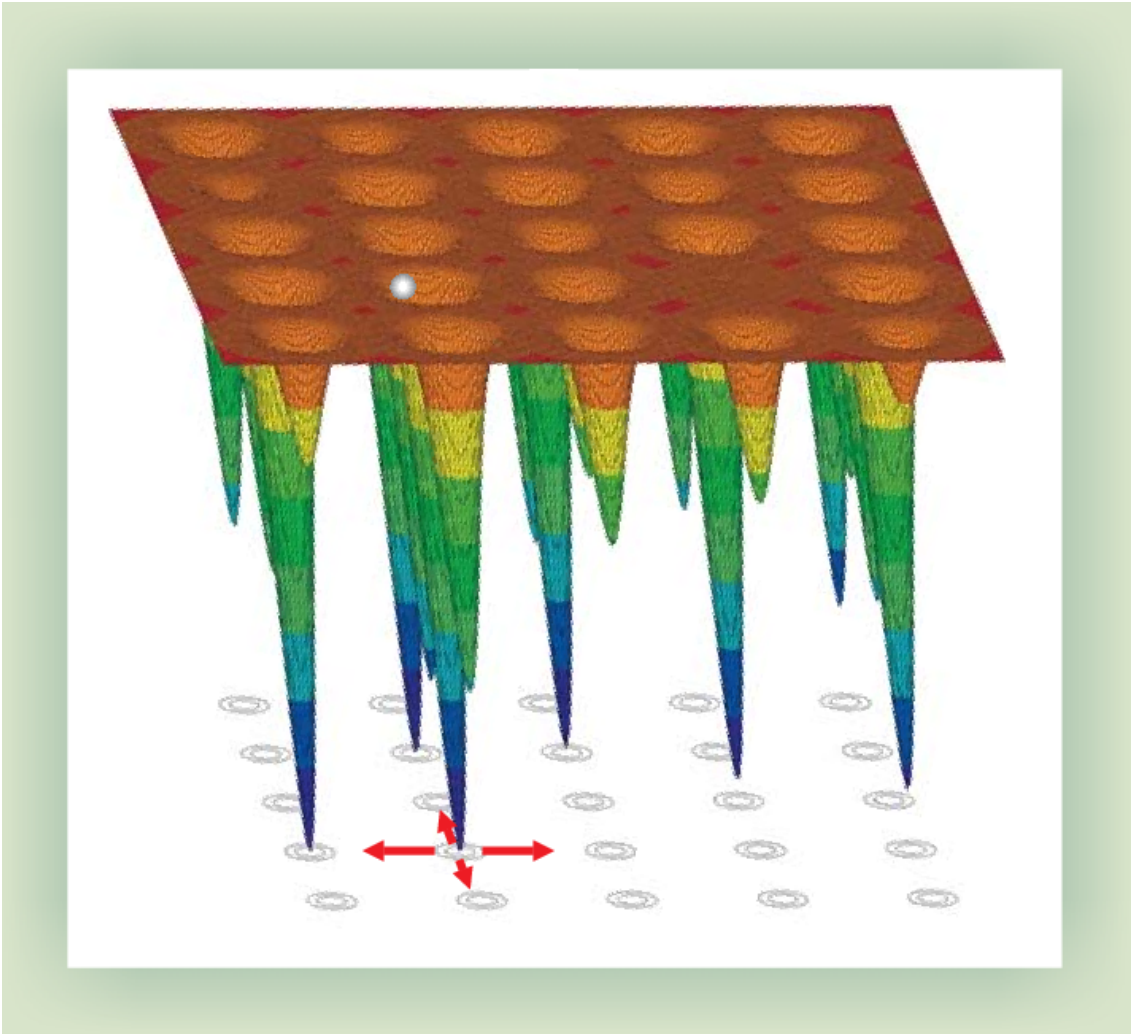


Figure: The Anderson model[1].

[1] Picture taken from *Fifty Years of Anderson Localization*

The Anderson Model

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$$H_t = \Delta + tV,$$

on the graph $G = \mathbb{Z}^d$.

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- Δ is the Laplacian matrix on G.
- V is a random diagonal matrix with $V(j,j) \sim \mu$ i.i.d. for some probability distribution μ.

The Anderson Model

Quantum Mechanics:

- Possible wave functions of an electron $\iff$ eigenvectors of H_t;
- Corresponding energies $\iff$ eigenvalues of H_t.

Question

What can we say about the eigenvectors of H_t? Are they localized?

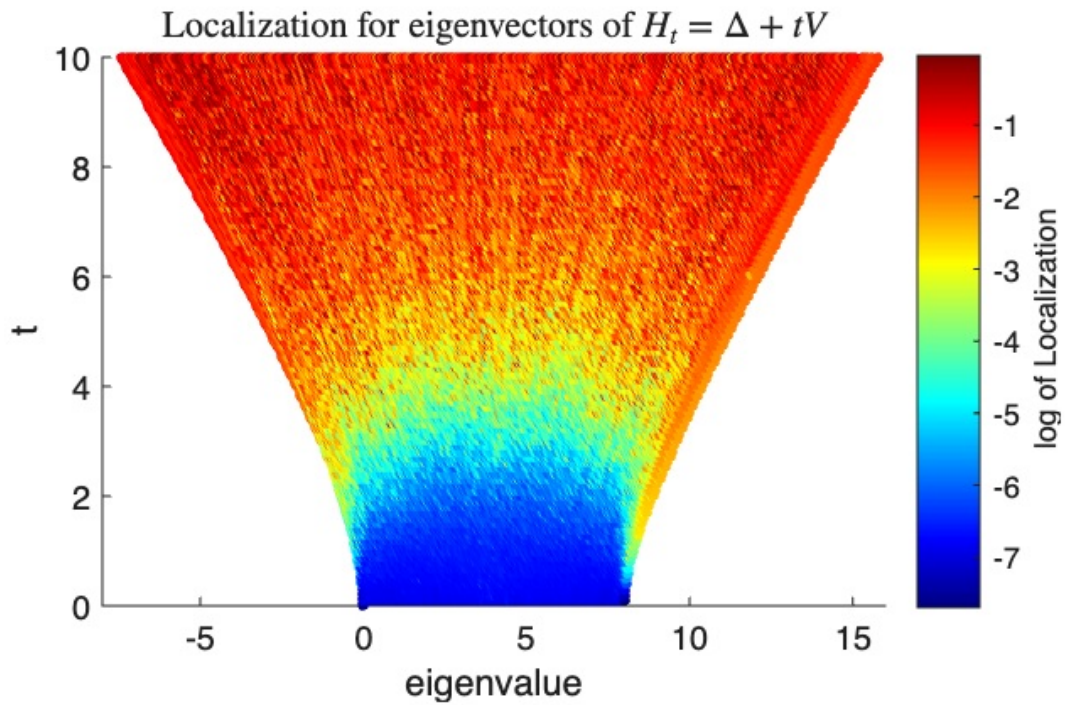


Figure: The localization–delocalization phase transition[2]

[2]Picture taken from Aizenman-Warzel

Known Results

Two kinds of distributions on V:

1. μ continuous (e.g. uniform on $[-1, 1]$);
2. μ discrete (e.g. -1 or 1 with probability $\frac{1}{2}$).

Known Results

Have proven things for

1. μ continuous and t large [Elgart–Klein];
2. μ discrete and $d \leq 3$ [Ding–Smart, Li–Zhang].

The Nodal Count

The nodal count $\nu(t, k)$ is the number of sign changes of the $k-$th eigenvector of H_t.

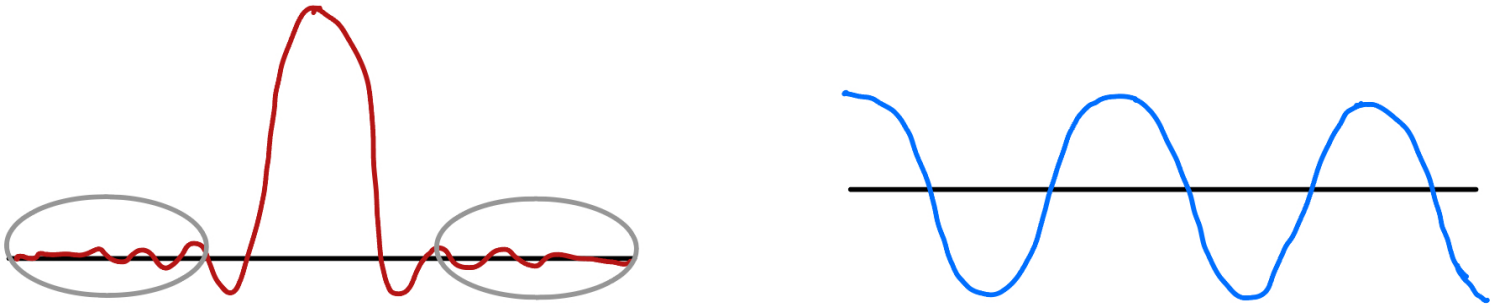


Figure: Can we determine localization from the variance of the nodal count?

The Nodal Count

We want finite nodal count, so we restrict:

$$G = (\mathbb{Z}/L_1\mathbb{Z}) \times \cdots \times (\mathbb{Z}/L_d\mathbb{Z}).$$

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Then,

$$(H_t\psi)(j) = \sum_{k\sim j}(\psi(j) - \psi(k)) + tv(j)\psi(j).$$

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Definition

H_t satisfies the Nodal Count Condition (NCC) if it has

- simple eigenvalues and
- nowhere vanishing eigenvectors.

Good and Bad Potentials

Lemma (NCC Dichotomy)

$H_t = \Delta + tV$ either fails to satisfy NCC for all $t \in \mathbb{R}$ or it satisfies NCC for all but finitely many t values.

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Definition

Call V bad if H_t fails to satisfy NCC for all $t \in \mathbb{R}$. Otherwise, call it good.

Our Results

When is V good/bad?

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Theorem (G.–Lindblad)

Let G be any connected graph. If μ is absolutely continuous with respect to the Lebesgue measure, then $\mathbb{P}(V \text{ is good}) = 1$.

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Suppose μ assigns 1 with probability p and -1 with probability $1-p$.

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For any d and any $L_1, \ldots, L_d$,

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If $d = 1$ and the side length L is prime, then

$$\mathbb{P}(V \text{ is bad}) = L(p^2 + (1-p)^2)^{\frac{L-1}{2}} - (L-1)(p^L + (1-p)^L).$$

Continuous Distributions

One-parameter Perturbations

Our work stems from the following result:

Lemma (Kato–Rellich)

A one-parameter family of normal matrices H_t that depends analytically on t admits an orthonormal basis of eigenvectors that depends analytically on t. Furthermore, their corresponding eigenvalues also depend analytically on t.

One-parameter Perturbations

Although we are considering $\Delta + tV$, we can afford to look at

- $s\Delta + (1-s)V$ or
- $V + t\Delta$

instead.

Algebraic Independence

Proposition

Suppose V is a diagonal matrix with algebraically independent diagonal entries. Then, V is good.

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- Instead of $\Delta + tV$, we will look at $V + t\Delta$.
- It suffices to find one t such that $V + t\Delta$ satisfies NCC.

Support of Eigenvectors

Let $\varphi_{k,t}$ be the $k-$th eigenvector of $V + t\Delta$.

$$\varphi_{k,t} = e_k + \sum_{j=1}^{\infty} t^j \varphi_k^{(j)}.$$

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Want: for all k and all vertices i, there exists j such that

$$\varphi_k^{(j)}(i) \neq 0.$$

Support of Eigenvectors

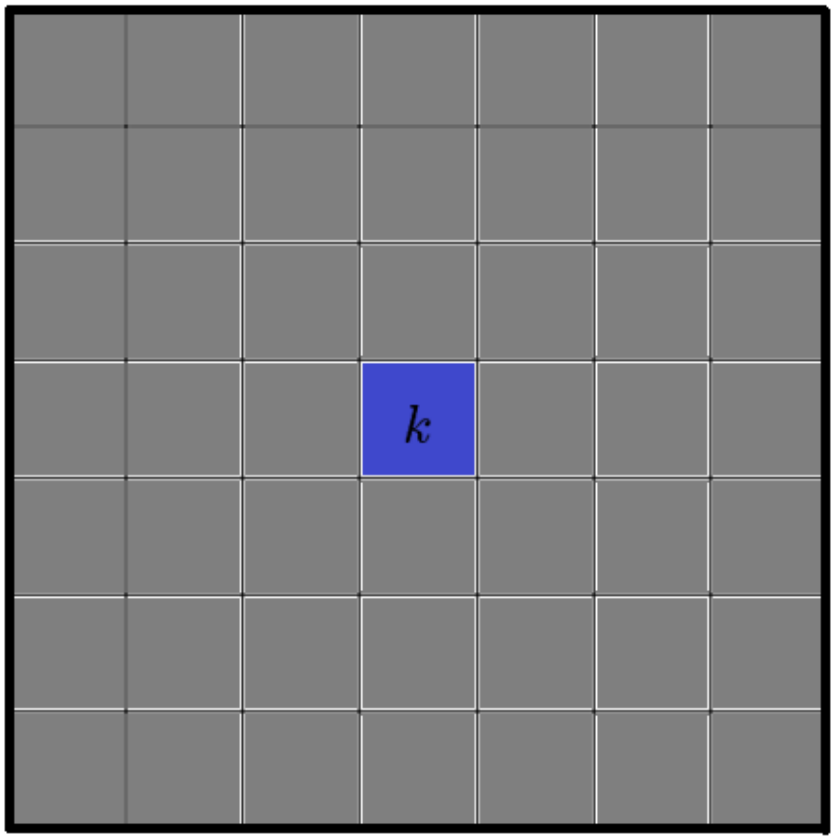


Figure: Support of $\varphi_k^{(0)}$

Support of Eigenvectors

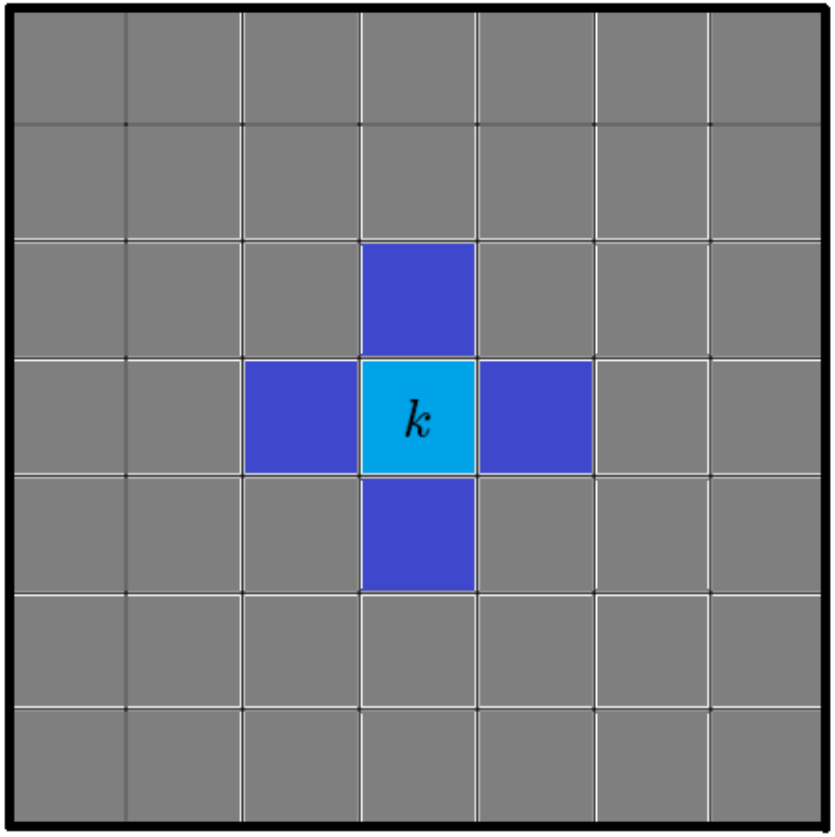


Figure: Support of $\varphi_k^{(1)}$

Support of Eigenvectors

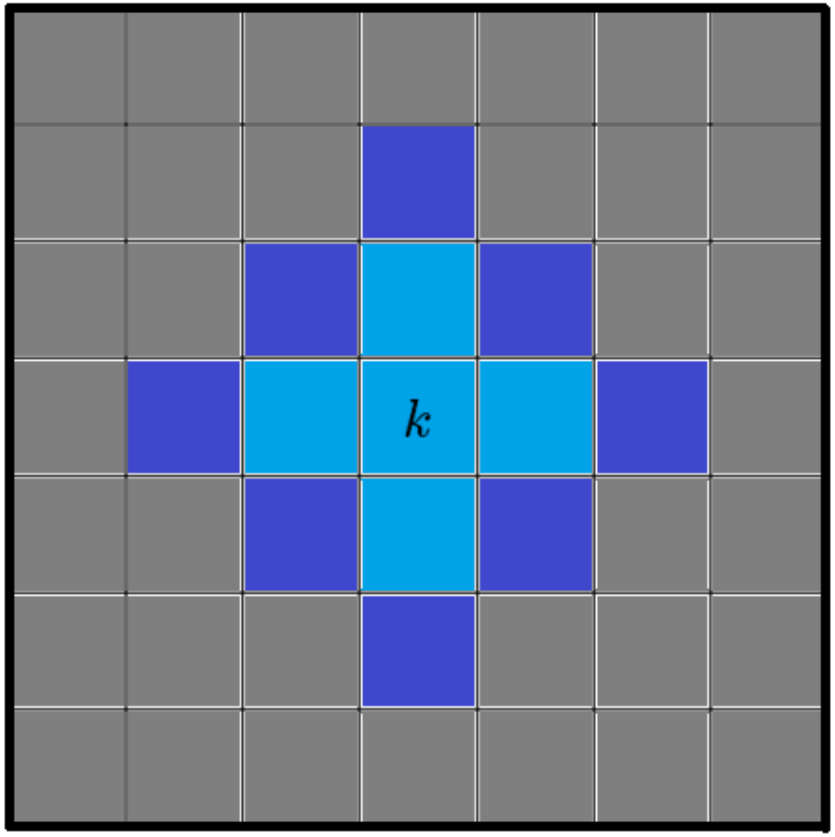


Figure: Support of $\varphi_k^{(2)}$

Support of Eigenvectors

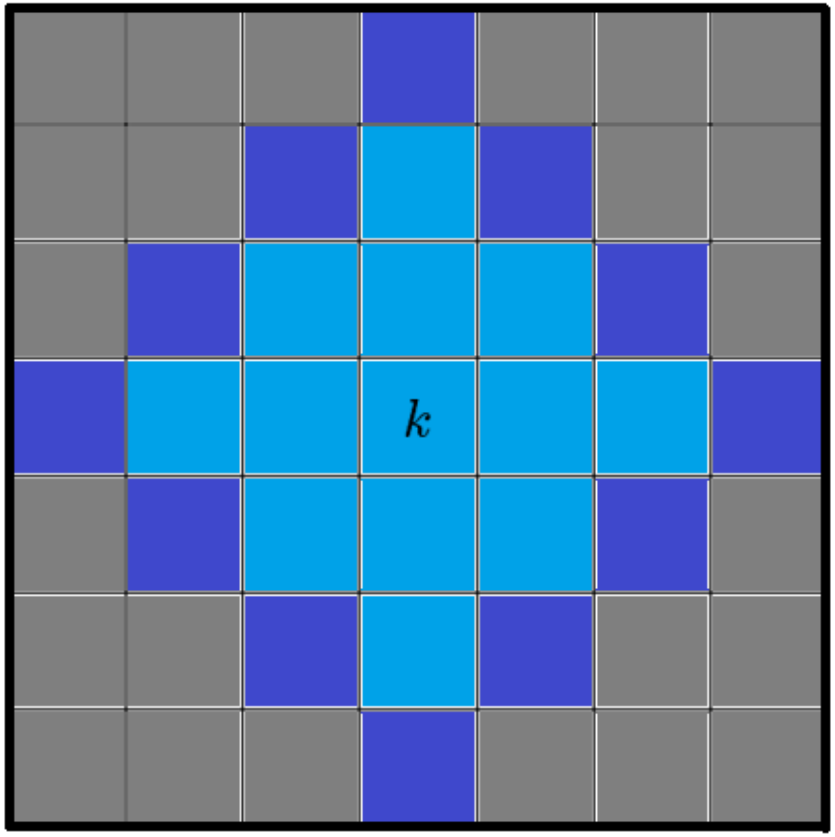


Figure: Support of $\varphi_k^{(3)}$

Support of Eigenvectors

Lemma

$$\varphi_k^{(j)}(i) = \begin{cases} 0 & \text{if } \operatorname{dist}(i,k) > j \\ \text{nonzero} & \text{if } \operatorname{dist}(i,k) = j \end{cases}.$$

Proof of Theorem 1

Theorem (G.–Lindblad)

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- If V is bad, our proposition implies its entries are algebraically dependent.
- μ gives probability 0 to any set with Lebesgue measure 0.



Discrete Distributions

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Answer:

No! Take $V = \pm \mathrm{I}$.

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What might be causing V to be bad?

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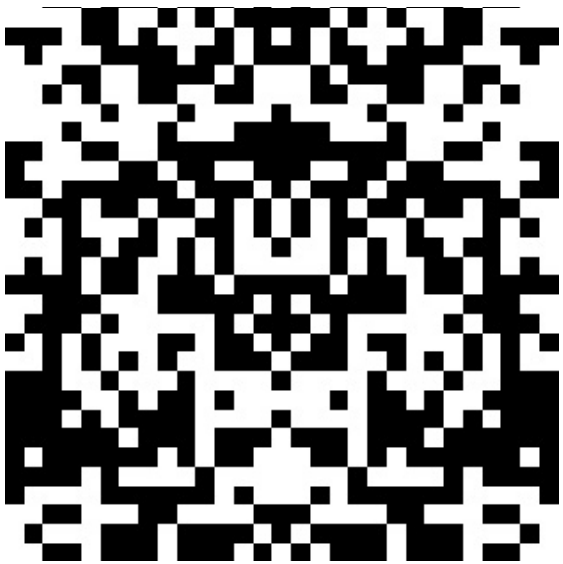
Out of the following three potentials, two are bad. Which ones?

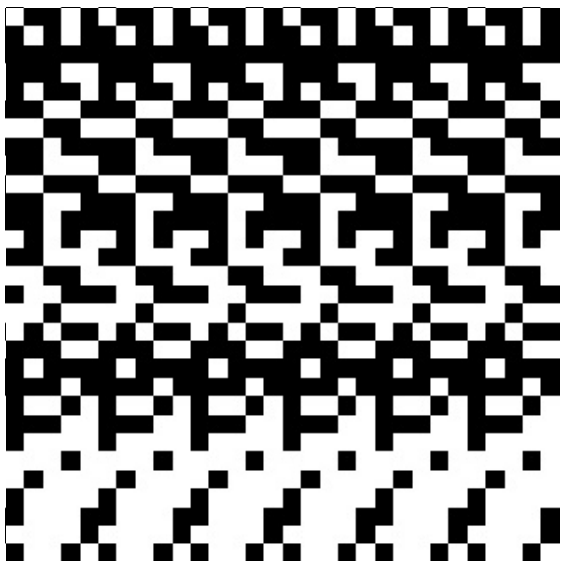
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Symmetries

Proposition

If both Δ and V commute with an orthogonal matrix $\mathcal{O} \neq \mathrm{I}$ that satisfies one of the following:

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Note: If G has an odd number of vertices, then any shared permutation matrix causes NCC failure for all t.

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- If $\mathcal{O}e_j = e_j$, then

$$(\lambda_{k,t} - 1)\langle \varphi_{k,t}, e_j \rangle = \langle \mathcal{O}\varphi_{k,t}, \mathcal{O}e_j \rangle - \langle \varphi_{k,t}, e_j \rangle = 0.$$

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- $\mathcal{O} \neq \mathrm{I}$ means we can choose $\lambda_{k,t} \neq 1$, giving $\varphi_{k,t}(j) = 0$.



Proof of Theorem 2

Theorem (G.–Lindblad)

For any d and any $L_1, \dots, L_d$,

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Proof.

- G has translation symmetry $(r_1, \ldots, r_i, \ldots, r_d) \mapsto (r_1, \ldots, r_i + 1, \ldots, r_d)$ for all i.
- Count the number of potentials with some translation symmetry.



Converse?

Is it true that if V is bad, we can find some nontrivial commuting orthogonal matrix $\mathcal{O}$?

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Is it true that if V is bad, we can find some nontrivial commuting orthogonal matrix $\mathcal{O}$?
In general, the answer is no! However, there are some cases when we can.

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Claim:

For $d = 1$, L an odd prime, V is bad if and only if V has a reflection symmetry about a vertex.

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To prove this, we use the Discrete Fourier Transform and the $L-$th roots of unity.

Prime Cycles

Proof.

Let $\omega = \exp(2\pi i/L)$. The *Discrete Fourier Transform* is the unitary matrix

$$\mathcal{F}(r,s) := \frac{1}{\sqrt{L}}\omega^{rs}$$

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Define

$$P(z) := \frac{1}{L}\sum_{j=0}^{L-1}v(j)z^j$$

Prime Cycles

Proof.

- In this framework, H_t degenerate for all t implies the matrix

$$\begin{pmatrix} P(1) & P(\omega^{2k}) \\ P(\omega^{-2k}) & P(1) \end{pmatrix}$$

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- The minimal polynomial of ω^{2k} is $\Phi = \sum_{j=0}^{L-1} z^j$, which forces $V = \pm I$.

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- Since $f(z) := \sum_{r=0}^{L-1} (v(j+r) - v(j-r))z^r$ has constant coefficient zero, it must be the zero polynomial.



Proof of Theorem 3

Theorem (G.–Lindblad)

If $d = 1$ and the side length L is prime, then

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Note that if $0 < p < 1$, then

$$\lim_{L \to \infty} \mathbb{P}(V \text{ is bad}) = 0.$$

Summary

We proved:

1. The NCC is generic for continuous distributions;
2. Symmetries cause potentials to be bad, causing NCC to fail with positive probability for discrete distributions.
3. For prime cycles, all bad potentials have symmetry. Moreover, the probability to be bad vanishes as we take the side length to infinity.